

Integrative Case Study: Non-coding Variant Interpretation

Session 6

T6: Making the most of cis-regulatory annotations for genome-scale analysis

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03/09/2026

Recap

Earlier sessions introduced the following components:

- ✓ *Session 1*: What cis-regulatory annotations are and why they matter.
- ✓ *Session 2* and *Session 3*: How major resources define and provide regulatory annotations and supporting evidence.
- ✓ *Session 4*: How to work with and analyze different annotation and evidence files in R and Python.
- ✓ *Session 5*: How to inspect tracks and annotations in IGV.

This session **integrates all the previous sessions in a realistic analysis workflow for interpreting non-coding variants.**

Format

- ✓ Students are grouped into 4-5 working groups, ideally up to 5 members per team.
- ✓ Two case studies are presented.
- ✓ Each group is assigned to one of the two case studies.
- ✓ Attendees have ~45 minutes for working in the hands-on.
- ✓ Each group select one member that share the results to the class in the discussion round.
- ✓ Wrap-up and closing remarks.

Case study 1: Human disease locus MYC / 8q24

- ✓ The 8q24 cancer risk variants [rs6983267](#) and [rs67491583](#) show long-range interaction with [MYC](#) in colorectal cancer ^{1,2}.
- ✓ MYC is one of the most important oncogenes in cancer³:
 - It is activated in the vast majority of cancers by genetic, epigenetic or post-translational mechanisms.
 - Its activation drives cancer progression through mechanisms involving either the cell-intrinsic acquisition of hallmarks of cancer or dysregulation of the tumour microenvironment and host immune responses.
 - Currently, no direct inhibitors of MYC are approved; however, many therapeutic agents targeting MYC are under development.

¹Pomerantz, Mark M et al. "The 8q24 cancer risk variant rs6983267 shows long-range interaction with MYC in colorectal cancer." Nature genetics vol. 41,8 (2009): 882-4. doi:10.1038/ng.403

²Tuupanen, Sari et al. "Characterization of the colorectal cancer-associated enhancer MYC-335 at 8q24: the role of rs67491583." Cancer genetics vol. 205,1-2 (2012): 25-33. doi:10.1016/j.cancergen.2012.01.005

³Dhanasekaran, Renumathy et al. "The MYC oncogene — the grand orchestrator of cancer growth and immune evasion." Nature Reviews Clinical Oncology vol. 19, 23–36 (2022). doi:10.1038/s41571-021-00549-2

Case study 1: Human disease locus MYC / 8q24

- ✓ Main question: *Do these 8q24 non-coding variants overlap regulatory elements that could plausibly influence MYC in a colorectal-relevant context?*
- ✓ Guiding questions:
 - How far are the variants from the MYC gene? Could it be possible that these variants belong to a regulatory region?
 - Do the CREs containing rs6983267 and rs67491583 show tissue- or context-specific enhancer activity?
 - [HCT116](#) and [Caco-2](#) are colorectal cancer cell lines. Are they included in the studied resources? If yes, which epigenomic evidence is available for each of them?
 - How is the regulatory activity in the locus for HCT116 and Caco-2?
 - Do the CREs annotation resources show agreement in the studied locus?

Case study 2: Mouse Dlx5/Dlx6 and the I56i enhancer

- ✓ The [Dlx5/Dlx6](#) intergenic region contains the conserved enhancer I56i (GRCm39/mm10, chr6:6,869,098-6,869,533), which has been described as a forebrain enhancer active in GABAergic interneuron development ⁴.
- ✓ chr6:6,869,492 A>G in I56i was reported to impair enhancer activity in the developing forebrain and adult cortex ⁴.
- ✓ The location of the SNP coincides with one of the two putative Dlx binding sites within I56i. This region of I56i also encodes the second exon of the long polyadenylated non-coding RNA (lncRNA) Evf2 (Dlx6os1).

⁴Poitras, Luc, et al. "An SNP in an ultraconserved regulatory element affects Dlx5/Dlx6 regulation in the forebrain." *Development* 137.18 (2010): 3089-3097.

Case study 2: Mouse Dlx5/Dlx6 and the I56i enhancer

- ✓ Main question: *Does the I56i candidate sequence correspond to a current regulatory annotation near Dlx5/Dlx6? How strong is the evidence that it acts as a forebrain enhancer?*
- ✓ Guiding questions:
 - Are there regulatory elements in the Dlx5/Dlx6 locus?
 - Is the region strongly regulated in forebrain developmental stages?
 - Do the epigenomic evidence tracks support the I56i enhancer region?
 - Do the CREs annotation resources show agreement in the studied locus?

Hands-on!



Discuss your team's findings



➤ What is the genomic context of the candidate region?

Where does it lie relative to nearby genes, promoters and annotated regulatory features, and could it plausibly regulate a gene over a long genomic distance?

➤ Is the evidence relevant to the biological context?

Which cell types, tissues or developmental stages provide the most relevant epigenomic evidence, and is activity specific to those contexts?

➤ What evidence supports its CRE annotation?

Do chromatin accessibility, histone-modification, transcription-factor-binding, conservation and regulatory-annotation tracks agree on whether the region is likely to be an enhancer or another type of CRE?

➤ Bonus: How strong is the conclusion, and how could it be tested?

What can be concluded from the available data, what remains uncertain, and which functional experiment would best test the proposed regulatory effect?